Police-Reported Family and Non-Family Violence in Canada, 2009–2024

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***Abstract*—This project analyzes police-reported family and non-family violence in Canada from 2009 to 2024 using Statistics Canada Uniform Crime Reporting data. The study uses exploratory data analysis to describe long-term changes in victim counts by year, gender, age group, and relationship of the accused to the victim. It then applies a log-transformed linear regression model to examine whether the annual trend in family-violence victim counts differs by gender, and uses simulation-based inference to compare mean victim counts between family and non-family violence categories. The expected contribution is a reproducible IEEE-style analysis that identifies major victimization patterns and demonstrates how visualization, regression, and resampling methods can be used to study public safety data.**

***Index Terms*—Family violence, non-family violence, police-reported victimization, exploratory data analysis, regression, simulation-based inference.**

I. Introduction

Violence remains an important public safety and public health issue in Canada. Family violence is especially important because it often occurs within close personal relationships, can involve repeated harm, and may be underreported. Non-family violence is also important because it represents a large share of police-reported violent victimization and may follow different demographic patterns. This project analyzes police-reported family and non-family violence in Canada from 2009 to 2024 using a Statistics Canada data table drawn from the Uniform Crime Reporting Survey [1]. Statistics Canada also reports continuing concern about family violence trends in recent years [2].

The main purpose of this project is to convert a previous group-project structure into an individual AMOD 5430 project completed by Helen Wang. The report uses an IEEE-style Quarto format, hides the source code in the rendered PDF, and presents the analysis through text, figures, and tables. Although the uploaded Excel file is named `2019-2024.xlsx`, the actual `REF_DATE` field contains observations from 2009 to 2024; therefore, the analysis uses the full available period.

This project is guided by two research questions. First, how did police-reported victim counts change over time, and do family-violence trends differ by gender? Second, is there a statistically meaningful difference in mean victim counts between family and non-family violence categories? These questions are addressed using exploratory data analysis, data visualization, regression modeling, and simulation-based inference.

II. Data and Methodology

The data set contains aggregated police-reported victim counts by year, gender of victim, age group of victim, and relationship of the accused to the victim. The analysis keeps Canada-level observations, the statistic labelled as number of victims, two gender groups, selected age categories, and the broad family and non-family relationship categories. After cleaning, the working data set contains 320 rows from 2009 to 2024, representing a total of 11,007,498 victim-count records across the selected categories.

The methodology has three parts. First, exploratory data analysis is used to summarize annual trends, relationship-category shares, gender differences, age-group patterns, and the distribution of count values. Second, a multiple linear regression model is fitted to family-violence victim counts using year, gender, and their interaction as predictors. Because victim counts are right-skewed and the spread of residuals can increase with the fitted value, the final model uses the natural logarithm of total family-violence victims as the response variable. Third, a permutation test and bootstrap confidence interval are used to compare mean victim counts between family and non-family violence categories without relying strongly on normality assumptions.

Table I: Cleaned data summary

rows	first_year	last_year	total_victims	min_count	max_count
320	2009	2024	11007498	5612	151741

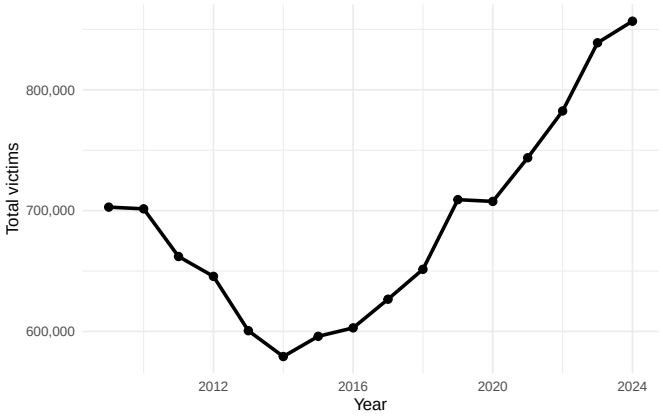


Figure 1: Overall trend of police-reported victims in Canada, 2009–2024.

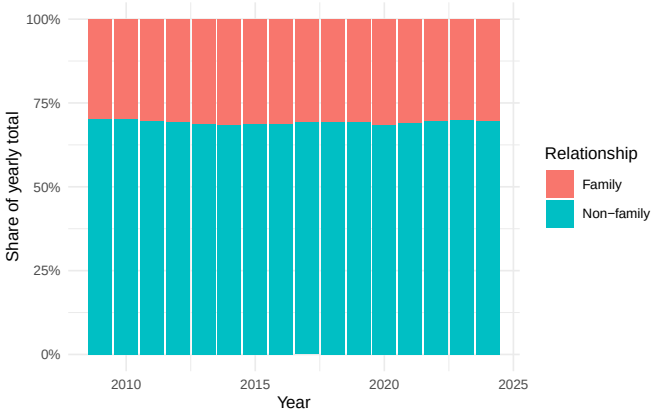


Figure 2: Family and non-family violence share by year.

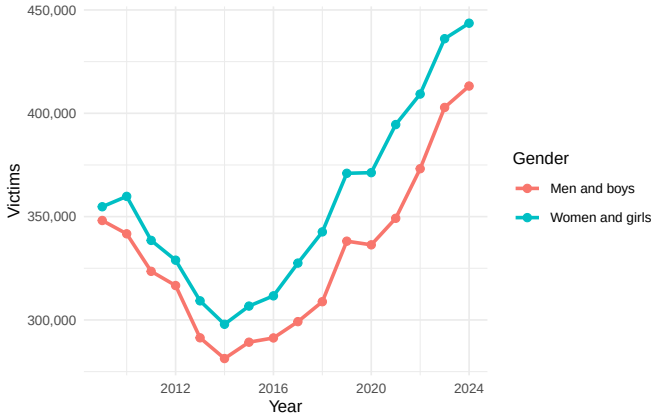


Figure 3: Victim counts by gender over time.

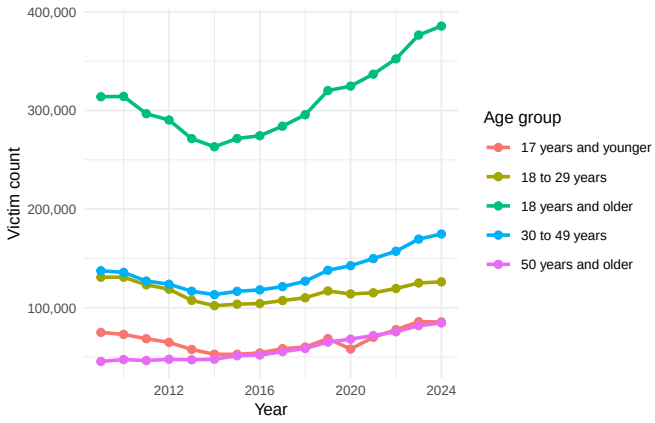


Figure 4: Victim counts by age group, 2009–2024.

III. Exploratory Data Analysis

A. Overall Annual Trend

Figure 1 shows the total number of selected police-reported victim-count records by year. The total increased from 702,900 in 2009 to 856,768 in 2024, an overall increase of approximately 21.9%. This indicates a clear long-term upward movement in the selected police-reported victim-count categories.

B. Family and Non-Family Violence Share

Figure 2 compares the annual share of family and non-family violence. Across the study period, family violence accounts for an average of approximately 30.7% of the selected yearly victim-count total, while non-family violence accounts for approximately 69.3%. The proportions are relatively stable over time, which suggests that the increase in total victim counts is not driven by only one relationship category.

C. Gender-Based Trends

Figure 3 presents the annual victim counts by gender. The figure allows visual comparison of the level and slope of victimization for men and boys versus women and girls. The gender trajectories appear to move upward over time, but the difference in levels indicates that gender should be included as a key explanatory variable in the regression model.

D. Age Group Trends

Figure 4 describes victim-count patterns across age groups. The broad adult category, `18 years and older`, is not mutually exclusive with narrower adult categories such as `18 to 29 years`, `30 to 49 years`, and `50 years and older`. Therefore, this plot should be interpreted as a descriptive comparison of table categories rather than a decomposition into fully separate age groups.

E. Indexed Growth Comparison

Figure 5 converts each gender-by-relationship series to an index with 2009 equal to 100. This view separates proportional growth from differences in absolute scale, making it easier to compare whether groups changed at similar rates over the study period.

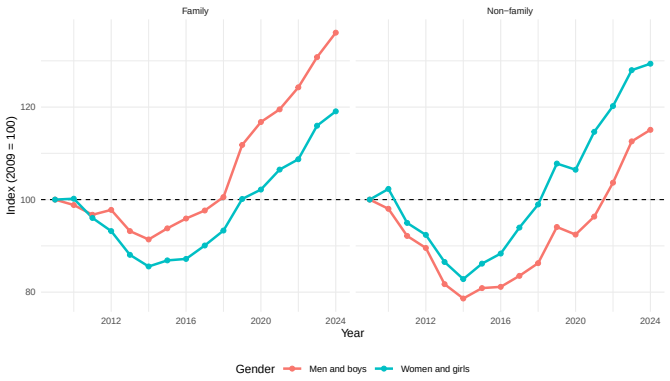


Figure 5: Indexed victim-count trends by gender and relationship category (2009 = 100).

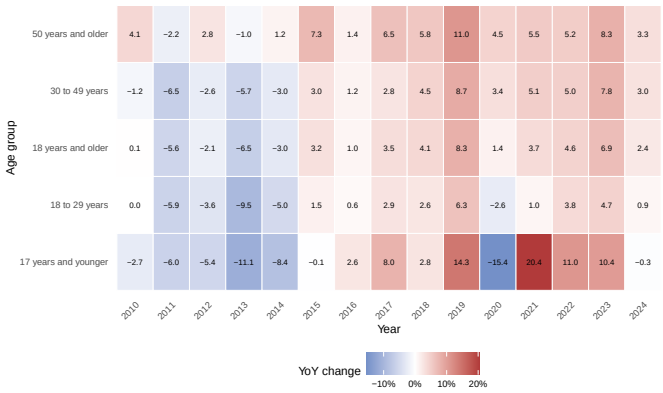


Figure 6: Heatmap of year-over-year percentage change in victim counts by age group.

F. Year-over-Year Change Heatmap

Figure 6 summarizes annual percentage changes by age category. The heatmap emphasizes the timing and direction of change rather than the size of each category, allowing periods of broad decline or acceleration to be identified quickly.

G. Distribution and High-Count Cells

Figure 7 shows that count values are right-skewed, especially because several aggregated cells contain very large victim counts. This distribution supports the use of a log transformation for regression and simulation-based inference for group comparison.

Table 2 lists the top ten highest count cells. These cells help identify which combinations of year, gender, age group, and relationship category contribute most strongly to the upper tail of the count distribution.

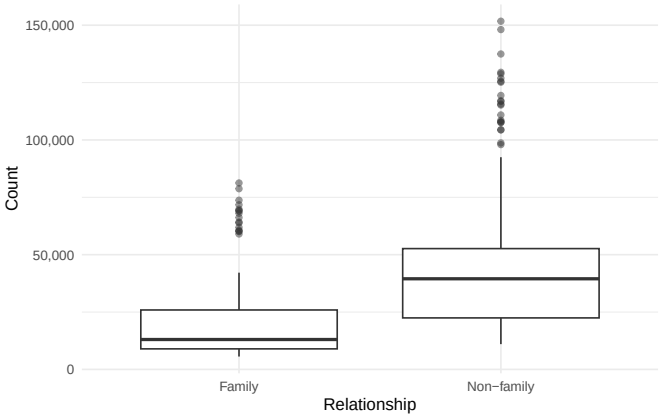


Figure 7: Distribution of cell counts by relationship type.

Table II: Top 10 highest Year by Gender by Age Group by Relationship cells

Year	Gender	Age group	Relationship	Count
2024	Men and boys	18 years and older	Non-family	151741
2023	Men and boys	18 years and older	Non-family	148093
2022	Men and boys	18 years and older	Non-family	137454
2021	Men and boys	18 years and older	Non-family	129473
2009	Men and boys	18 years and older	Non-family	128795
2010	Men and boys	18 years and older	Non-family	126890
2020	Men and boys	18 years and older	Non-family	125544
2019	Men and boys	18 years and older	Non-family	125150
2011	Men and boys	18 years and older	Non-family	119389
2024	Women and girls	18 years and older	Non-family	117073

IV. Regression Analysis

The regression analysis focuses on family-violence victim counts. The predictor `year_centered` is defined as year minus 2009, so the intercept represents the estimated baseline for men and boys in 2009. The final model is:

$$\log(\text{family victims}) = \beta_0 + \beta_1(\text{year centered}) + \beta_2(\text{women and girls}) + \beta$$

The log transformation makes the coefficients easier to interpret as approximate percentage changes. It also helps stabilize the residual variance for count data.

Figure 8 overlays the observed family-violence totals with fitted trends from the log-transformed interaction model. The shaded confidence bands show uncertainty around the estimated mean trend and make the persistent difference between gender groups visually explicit.

Figure 9 shows the diagnostic plots for the raw-count model. Because the response variable is an aggregated count, the residual pattern may show unequal variance. For this reason, the final interpretation is based on the log-transformed model.

Table 3 reports the coefficients from the log-transformed regression model. The coefficient for year indicates an

Table III: Estimated coefficients of the log-transformed regression model

[!t]

Term	Estimate	Std. Error	t value	p-value
Baseline: Men and boys in 2009	10.88121	0.03728	291.88735	0.00000
Annual growth rate for Men and boys	0.02300	0.00423	5.43183	0.00001
Baseline difference: Women and girls	0.91161	0.05272	17.29148	0.00000
Difference in annual growth rate	-0.00953	0.00599	-1.59108	0.12282

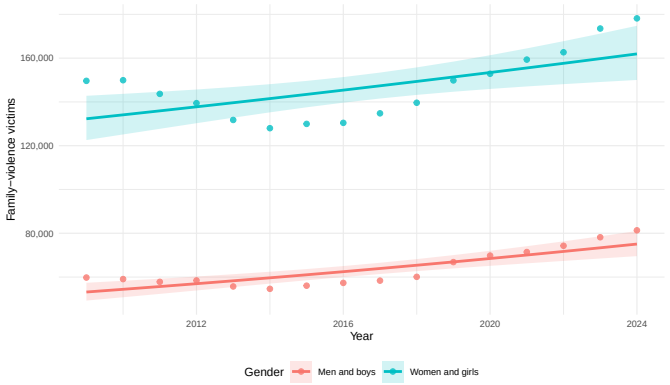


Figure 8: Observed and model-predicted family-violence victim counts with 95% confidence bands.

Table IV: Model fit statistics

[!t]

r.squared	adj.r.squared	statistic	p.value
0.97183	0.96881	321.9392	0

estimated annual change of approximately 2.3% for the reference group, men and boys. The gender coefficient implies that the baseline expected family-violence victim count for women and girls is approximately 2.49 times the count for men and boys, holding the baseline year constant. The interaction p-value is 0.123, so the evidence for different annual growth rates by gender is limited under this model.

Figure 10 presents the diagnostic plots after log transformation. Compared with the raw-count model, the log model is more appropriate for interpretation because it reduces the influence of large count values and makes the residual spread more stable.

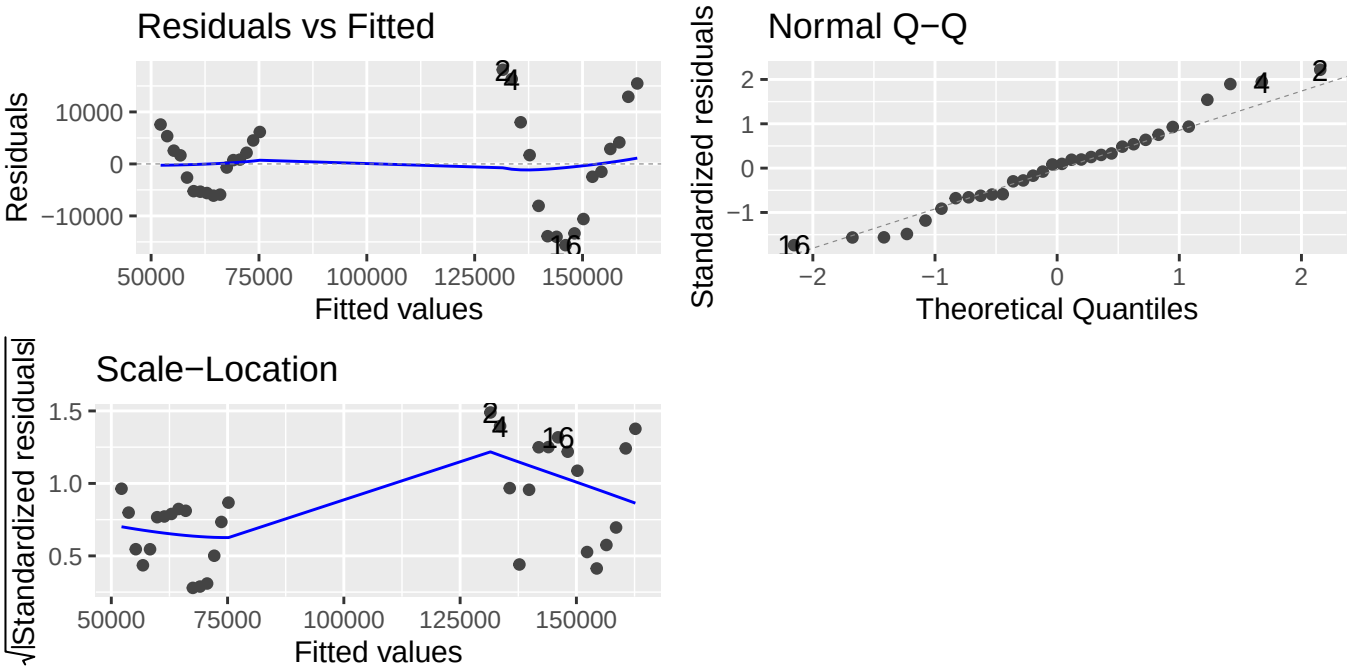


Figure 9: Diagnostic plots for the raw-count regression model.

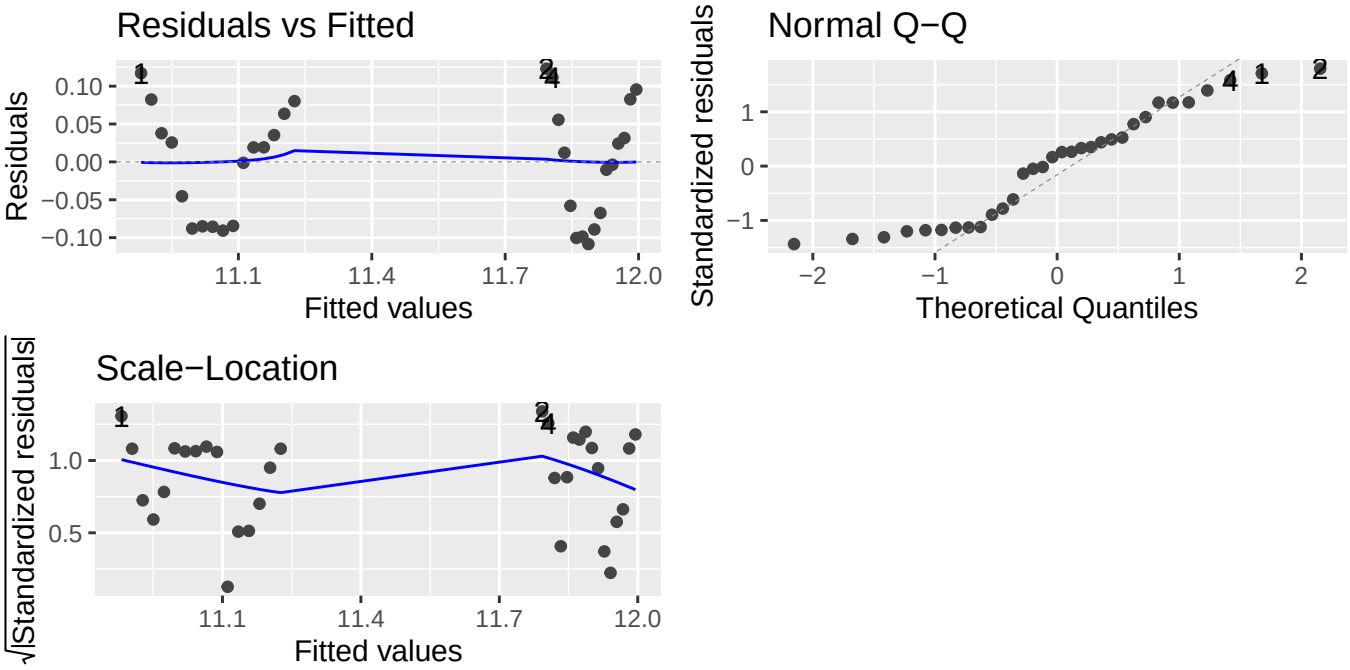


Figure 10: Diagnostic plots for the log-transformed regression model.

V. Simulation-Based Inference

The second research question compares mean victim counts between non-family and family violence cells. The observed mean difference is calculated as non-family minus family. The observed difference is 26,637, meaning that the selected non-family violence cells have a higher mean victim count than the selected family violence cells in the sample.

A permutation test is used to assess whether this difference could plausibly occur if relationship category and count were independent. A bootstrap procedure is also used to estimate the uncertainty around the difference in means.

Table V: Simulation-based inference summary

[!t]

Observed mean difference	Permutation p-value	Bootstrap CI lower	Bootstrap CI upper
26637.26	0	20696.3	32339.67

Figure 11 displays the null distribution from the permutation test. The observed difference lies far outside the main body of the distribution, providing a direct visual explanation for the very small p-value.

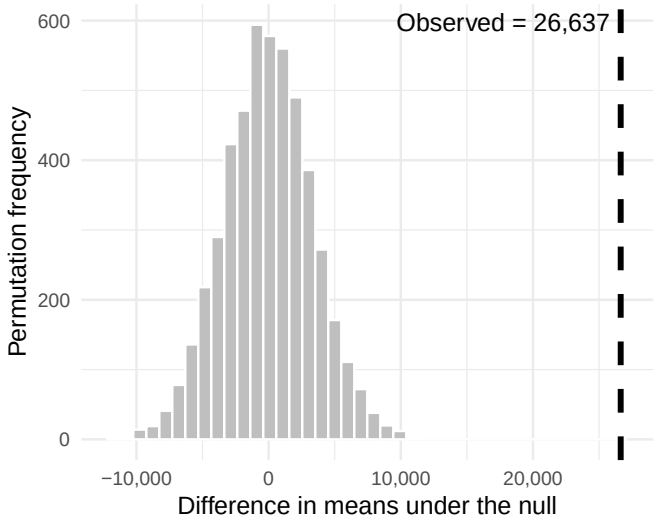


Figure 11: Permutation null distribution for the difference in mean victim counts.

The permutation p-value is 0. The 95% bootstrap confidence interval for the mean difference is [20,696, 32,340]. Because this interval is positive, the analysis supports the conclusion that the mean victim count is higher for non-family violence cells than for family violence cells in the selected data.

VI. Discussion

The exploratory analysis shows a long-term increase in police-reported victim counts from 2009 to 2024. The relationship-share plot suggests that family and non-family violence both remain important categories across the full period. The gender trend plot provides motivation for modeling gender as a key predictor, while the distribution plot shows why transformations and resampling methods are useful for this data set.

The regression results suggest that family-violence victim counts increased over time and that women and girls had higher baseline family-violence victim counts than men and boys. However, the interaction term does not provide strong evidence that the annual trend differs substantially by gender. In practical terms, the gender gap appears persistent rather than clearly narrowing over the study period.

The simulation-based comparison indicates that non-family violence cells have higher average victim counts than family violence cells. This does not mean family violence is less serious; rather, it reflects the structure and aggregation of the selected police-reported data. Family violence remains a policy-relevant issue because of its relationship context, possible repeat victimization, and underreporting concerns.

VII. Limitations

Several limitations should be considered. First, the data are aggregated counts rather than individual-level records, so the analysis cannot make claims about individual risk. Second, the data are police-reported, which means incidents not reported to police are not included. Third, some age categories overlap, especially the broad `18 years and older` group and the narrower adult groups; therefore, age-group figures should be interpreted descriptively. Fourth, the analysis is conducted at the Canada level and does not examine provincial or regional variation. Finally, the regression model is useful for summarizing linear trends, but it does not establish causality.

VIII. Conclusion

This project applies IEEE-style reproducible reporting to a Statistics Canada data set on police-reported family and non-family violence from 2009 to 2024. The analysis finds an overall increase in selected victim counts over time, persistent gender differences in family-violence victim counts, and higher average counts for non-family violence cells than family violence cells. The project also demonstrates the value of combining EDA, visualization, log-transformed regression, and simulation-based inference when working with skewed aggregated count data.

Future work could extend the analysis by using rates per population, separating provinces and territories, avoiding overlapping age categories, and comparing police-reported trends with victimization survey data. These extensions would provide a more complete understanding of violence patterns and reporting behaviour in Canada.

References

[1] Statistics Canada, “Victims of police-reported family and non-family violence, by gender and age group of victim, and detailed relationship of accused to victim.” Data Table, Catalogue No. 35-10-0067-01, 2024. Available: https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3510006701

[2] Statistics Canada, “Police-reported family violence in canada, 2018 to 2023: An overview,” *Juristat*, vol. 45, no. 1, 2025, Available: https://www150.statcan.gc.ca/n1/pub/85-002-x/85-002-x2023001-eng.htm